# PHYSIOLOGIC EFFECTS OF MASSAGE

## CLINICAL & ACADEMIC REFERENCE MANUAL

**Definition:** Mechanical stimulation of tissues by rhythmically applied pressure and stretching.

**Primary Clinical Applications:** Often used to increase flexibility and coordination, decrease pain & neuromuscular excitability, stimulate circulation, and facilitate healing.

The physiological and clinical effects of massage therapy are broad and multi-systemic. To establish a structured clinical framework, the effects of massage may be broadly classified as either **reflexive** or **mechanical**.

## Core Classifications of Massage Effects

### *Reflexive Effects*

Nervous system response. Effects sensory and motor nerves locally and some central nervous system pathways.

### *Mechanical Effects*

Makes mechanical or histological changes in myofascial structures through direct force applied superficially.

## Reflexive Effects: Mechanisms and Sub-types

### *General Reflexive Mechanisms*

- Attempts to exert effects through skin and superficial connective tissues.
- Reflex mechanism is believed to be an autonomic nervous system phenomenon.
- Contact stimulates cutaneous receptors.
- Causes sedation, relieves tension, and increases blood flow.

### *Effects on Pain*

- Modulates pain through the gate control theory & ***β***-endorphins.

### *Effects on Circulation*

- Light touch causes transient dilation of lymphatics and small capillaries.
- Increase blood flow.
- Increased temperature in the treated area.
- Increased lymphatic flow.
- Assists in the removal of edema.
- May impact lactate clearance.

### *Effects on Metabolism & Cardiovascular Function*

- Does not alter general metabolism.
- No changes in the acid-base equilibrium of blood.
- No significant systemic cardiovascular effects.
- Assists specifically in the removal of lactic acid.

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## Mechanical Effects: Deep Tissue Alterations

Mechanical effects stretch muscle, elongate fascia, and mobilize adhesions/restrictions. These actions are *always accompanied by reflex effects*. However, a distinct inverse relationship exists: **As mechanical stimulus increases → reflex effect decreases.** Targets deep tissues including muscle, tendon, and fascia.

### *Effects on Muscle*

- Stretch intramuscular connective tissue.
- Trigger points relief.
- Increase blood flow to skeletal muscle.
- Retard muscle atrophy after injury.
- Increase Range of Motion (ROM).
- **Does NOT** increase strength or muscle tone.

### *Effects on Skin*

- Increase skin temperature.
- Increase sweating.
- Loosen adhesions and soften scar tissue.
- Break fibrous scar tissue.
- Break adhesions between skin & subcutaneous tissue.

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## Psychological Effects & Treatment Considerations

### *Psychological Effects*

- Psychological benefits are as beneficial as physiological effects.
- “Hands on” effect improves the patient's perception of care.
- Lowers psycho-emotional & somatic arousal.
- Reduces tension & anxiety.

### *Clinical Knowledge Requirements*

- Knowledge of anatomy is essential.
- Thorough understanding of pathology is required.
- Thorough knowledge of massage principles is mandatory.

### *Positioning of the Clinician*

- Allows relaxation, prevents clinician fatigue, and permits free movement.
- Weight must be evenly distributed, with the ability to shift weight smoothly between feet.
- Fit hands perfectly to the body contour of the patient.
- Hands should always be warm prior to contact.

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## General Treatment Techniques & Guidelines

- **Pressure:** Depends entirely on tissue type and the patient's specific condition.
- **Rhythm:** Must be kept steady and even throughout the application.
- **Duration:** Depends directly on pathology, area size, speed, age, and patient condition.
- **Swelling Protocol:** Begin proximally to establish the "uncorking effect".
- **Pain Threshold:** Massage should never be painful (except where explicitly indicated like deep friction).
- **Direction of Strokes:** Apply parallel to the underlying muscle fibers.
- **Sequence:** Always begin and end the session with effleurage.
- **Patient Comfort:** Ensure the patient is warm, relaxed, and fully comfortable.
- **Lubricant:** Always use a proper lubricant to prevent friction injuries.
- **Stroke Pattern:** Ensure strokes overlap continuously.
- **Pressure Vector:** Pressure follows venous flow (return stroke direction).
- **Rhythmicity:** All strokes must remain highly rhythmic.

### *Equipment & Room Set Up*

- Treatment Table
- Linens and pillows for positioning and draping
- **Lubricant selection:** Should be slightly absorbed, but not slippery.
- **Optimal blend:** Beeswax : coconut oil (***1:3*** ratio)
- **Other acceptable lubricants:** Olive oil, mineral oil, cocoa butter, hydrolanolin, analgesic creams, alcohol, powder.

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## Classic Massage Techniques (Hoffa System)

The Hoffa system comprises four primary classifications of strokes: Effleurage, Petrissage, Tapotement, and Vibration.

### EFFLEURAGE (STROKING)

- Starts and ends every single massage session.
- Increases venous & lymphatic flow and increases skin circulation.
- Light pressure can be applied either centripetally or centrifugally.
- **Key Points:** Rhythmic stroking pattern; light pressure yields relaxation/reflex effects, deep pressure yields mechanical effects; must closely follow veins & lymph flow; palms used rhythmically; overlap strokes; keep hand contact continuous at all times.

### PETRISSAGE (KNEADING)

- Derived from the French word *"petrir"* meaning to knead.
- Involves the compression, shearing, and wringing of muscle tissue, utilizing a lift and glide motion.
- **Techniques:** Lift skin/subcutaneous tissue/muscle; knead firmly between thumb and fingers; wring the tissue like a dishrag.
- Often performed **without lubricant** to maximize tissue grip.
- **Primary Effects:** Breaks adhesions, stretches individual fibers, increases venous & lymphatic return, and removes metabolic waste.

### PERCUSSION (TAPOTEMENT)

- Consists of repeated, rhythmic striking of the soft tissues.
- **Types:** Tapping, pounding, cupping, hacking, and slapping.
- **Clinical Uses:** Respiratory care (to mobilize phlegm/secretions) and precompetition stimulation.
- **Technical Notes:**
  - *Hacking:* Performed using the ulnar side of the hand.
  - *Cupping:* Performed ensuring only the rim of the cupped hand makes contact.
  - *Raindrops:* Used specifically for relaxation and tissue desensitization.

### VIBRATION

- Involves shaking soft tissue rapidly over the underlying bone structure with minimal joint movement.
- **Clinical Uses:** Muscle relaxation or precompetition arousal.
- **Application Parameters:** Rapid vibration is used for precompetition arousal; moderate vibration is applied postcompetition. Can be applied manually or via machine.

## Advanced Specialized Clinical Modalities

### FRICTION MASSAGE

- Small circular movements targeted deep within the tissue layers.
- **Purpose:** Loosen adhesions, reduce muscle spasm, reduce localized edema, and produce distinct reflex effects. Mobilizes muscle, tendon, or scar tissue. Restarts the acute inflammation process when original healing is incomplete or chronic.
- **Technique:** Circular or transverse direction applied strictly across the fibers using deep pressure (the clinician's elbow may be utilized). This technique is inherently **painful**. Avoid completely in acute injuries. May be followed immediately with stretching.

## TRANSVERSE FRICTION

- Used explicitly to treat chronic tendon inflammation.
- Employs strong perpendicular pressure relative to the fiber direction.
- **Protocol:** Applied for 7–10 minutes every other day, aiming specifically to restart the healing process.

## CONNECTIVE TISSUE MASSAGE

- Specialized stroking within the connective tissue layers based on the principle that abnormal visceral or myofascial tension is reflected systemically.
- Causes significant local warmth & general relaxation. Performed with **no lubricant** while the patient is in a sitting position.
- **Protocol:** 15–25 minute sessions, conducted 2–3 times per week for a total of 15 sessions, followed by a mandatory rest period of 4 weeks.
- **Indications:** Scars, fractures, arthritis, low back pain, edema, Raynaud’s disease, claudication, migraines, Parkinson’s disease, gastrointestinal (GI) disorders, and reproductive disorders.

## TRIGGER POINT MASSAGE

- Targeted at hypersensitive points found within muscle, tendon, fascia, or periosteum. Pain is typically due to localized inflammation and presents with specific referred pain patterns.
- **Types:**
  - *Latent:* No spontaneous pain, but painful upon palpation.
  - *Active:* Causes pain at rest along with localized tenderness.
- **Clinical Signs:** Reduced Range of Motion (ROM), palpable hypersensitive nodules, and the elicitation of a positive "jump sign".
- **Treatment:** Application of direct sustained pressure or deep friction producing painful stimulation for 1–5 minutes per point, ultimately inducing a dulling or numbing effect.

## STRAIN-COUNTERSTRAIN

- A passive positioning technique where the patient is moved into a fully pain-free position.
- Reduces the overactive stretch reflex.
- **Protocol:** The shortened muscle position is held statically for **90 seconds** to induce complete neurological relaxation.

## POSITIONAL RELEASE THERAPY

- A modality very similar to strain-counterstrain.
- Differentiated by the explicit utilization of a facilitating force along with sustained compression on the identified tender point.

## ACTIVE RELEASE THERAPY (ART)

- Specifically designed to treat soft tissue adhesions.
- **Technique:** The clinician physically traps the affected muscle tissue while the patient actively moves the limb or joint through its range.
- **Protocol:** Repeated 3–5 times per session, and always followed immediately by targeted stretching & rehabilitation exercises.

## MYOFASCIAL RELEASE

- Direct soft tissue mobilization aimed specifically at releasing complex fascial restrictions across multi-plane restrictions.
- Improves overall ROM. Must be systematically followed by stretching, strengthening, and posture retraining.

## GRASTON TECHNIQUE

- An Instrument-Assisted Soft Tissue Mobilization (IASTM) approach utilizing specialized stainless steel tools.
- Used to precisely detect and treat soft tissue restrictions, break down dense scar tissue, and deliberately promote localized inflammation and healing. Always integrated comprehensively with active rehab modalities.

# MASSAGE TREATMENT TECHNIQUES

*Complete Academic & Clinical Unabridged Compendium*

## HOFFA MASSAGE

| TECHNIQUE CLASSIFICATION | CORE DESCRIPTION & COMPONENTS |
|---|---|
| **HOFFA MASSAGE** | Classical massage technique which uses a variety of superficial strokes:<br>• Effleurage<br>• Petrissage<br>• Tapotement<br>• Vibration |

## EFFLEURAGE (STROKING)

| CATEGORY / CONCEPT | FULL TECHNICAL SPECIFICATIONS & PRACTICAL RULES |
|---|---|
| **EFFLEURAGE (STROKING)** | • Every massage begins and ends with effleurage<br>• Increases venous and lymphatic flow<br>• Increases circulation to skin surface<br>• Start with a light pressure, move centripetally or centrifugally consistently throughout treatment |
| **EFFLEURAGE DEFINITION** | Gliding manipulation performed with light centripetal pressure that deforms subcutaneous tissue down to the deep fascia. Directed toward the heart or proximally. |
| **EFFLEURAGE (CONT.) Keys to Applying** | • Apply rhythmic stroking on the skin.<br>• Can be either light or deep.<br>• Light massage promotes relaxation and sensory reflexes.<br>• Deep massage promotes the mechanical effects.<br>• Increases blood and lymphatic circulation. |
| **Effleurage (cont.) N.B)** | *21-23/73*<br>Keep one hand in contact with skin at all times by stroking in one direction with both hands and return to beginning with light finger tip contact.<br>Or stagger hands, making contact with the second before lifting the first from the body.<br>Begin and end massage with light effleurage. |
| **EFFLEURAGE (CONT.) Stroke Types** | **Deep stroking:** Follow the course of veins and lymph vessels.<br>**Superficial stroking:** Follow the contour of body or the underlying tissue. Use palms of both hands in a deliberate, rhythmic fashion. |
| **EFFLEURAGE (STROKING) Mechanical Form** | Deep stroking is a form of effleurage, except it is given with more pressure to produce a mechanical effect. Kneading stroke is directed towards the heart. |

## PÉTRISSAGE (KNEADING)

| TECHNICAL ASPECT | FULL METHODOLOGICAL TEXT & CLINICAL PARAMETERS |
|---|---|
| **PÉTRISSAGE Etymology & Actions** | From the French *pétrir*, "to knead".<br>Group of related techniques that repetitively:<br>• Compress (squeeze)<br>• Shear (wring)<br>• Release muscle tissue<br>with varying amounts of drag. Lift and glide. |
| **PÉTRISSAGE (CONT.) Keys to Applying** | • Lift and knead the tissue (skin, subcutaneous, muscle).<br>• Lift tissue between thumb and fingers or fingers and palm.<br>• Then gently roll and knead tissue in hand.<br>• Gently wring muscle between two hands as if you were wringing water out of a dishrag.<br>• Often performed without lubricant or lotion.<br>• Stretches and separates muscle fiber, fascia, and scar tissue. |
| **PETRISSAGE (KNEADING) Manipulations** | Consists of kneading manipulations that press and roll muscles under fingers or hands.<br>• Muscles are gently squeezed.<br>• Lifted.<br>• Relaxed.<br>Hands may remain stationary or move along length of muscle or limb. |
| **PETRISSAGE (KNEADING) Purpose** | • To increase venous and lymphatic return.<br>• To press metabolic waste products out of affected areas through intensive vigorous action.<br>• Can also break up adhesions between skin and underlying tissue. |

## TAPOTEMENT (PERCUSSION)

| METHOD / RULE | FULL CLINICAL TECHNIQUES & GUIDELINES |
|---|---|
| **PERCUSSION (Tapotement)** | Repeated rhythmical light striking of the skin. Techniques include: Gentle tapping, Pounding, Cupping, Hacking, Slapping the skin. |
| **Percussion (cont.) Main Uses** | Two main uses:<br>• For respiratory ailments to promote phlegm mobilization.<br>• Stimulation for precompetition preparation. |
| **Percussion (cont.) Keys to Applying** | Hacking is done with the ulnar side of hand. With wrist and fingers limp. Karate chop. |
| **PERCUSSION (CONT.) Technical Rules** | During cupping, only the rim of the hand should come in contact with the body. |
| **PERCUSSION (CONT.) Nerve Variations** | **Raindrops Variation:** To promote relaxation and desensitization of irritated nerve endings.<br>Applied by lightly touching the skin with fingers in an alternating manner. Typing. ***33-36/73*** |
| **TAPOTMENT (PERCUSSION) System** | Uses a variety of percussive or beating techniques. Brisk blows administered with relaxed hands (rapid alternating movement). Used to: Increase circulation and blood flow, Stimulate peripheral nerve endings. |
| **TAPOTMENT (PERCUSSION) Forms** | Hacking, Slapping, Beating, Tapping, Clapping or cupping. |

## VIBRATION

| PARAMETER | FULL SYSTEM SPECIFICATIONS |
|---|---|
| **VIBRATION Description** | Shaking. Repetitively moving soft tissue back and forth over the underlying bone with joint motion minimal. |
| **VIBRATION (CONT.) Principal Uses** | • Relaxation of skeletal muscle.<br>• As a stimulus for precompetition and intercompetition owing to its effects of systemic arousal and enhanced awareness. |
| **VIBRATION (CONT.) Keys to Applying** | Apply moderate to rapid shaking strokes to the skin. Rapid for precompetition. Moderate for postcompetition. Can be applied with the hands or with a machine. |
| **VIBRATION Manual Execution** | A fine tremulous movement, made by hand or fingers placed firmly against a part causing a part to vibrate. Hands should remain in contact and a rhythmical trembling movement will come from arms. |

## FRICTION MASSAGE & TRANSVERSE FRICTION

| CLINICAL CONCEPT | FULL METHODOLOGICAL DESCRIPTION & STRUCTURAL RATIONALE |
|---|---|
| **FRICTION MASSAGE Purpose** | Loosen adherent fibrous tissue (scar). Aid in edema absorption. Reduce muscle spasm. Produce reflex effects. Involves small circular movements beneath superficial tissues directed at underlying structures. ***43-46/73*** |
| **FRICTION MASSAGE Rationale** | Rationale for applying friction massage is to mobilize muscle and separate adhesions in muscle, tendons, or scar tissue that restrict movement and cause pain. Sometimes chronic inflammation occurs because an injury does not go through the normal stages of inflammation and healing. ***45-47/73*** |
| **FRICTION MASSAGE Physiological Goal** | The purpose is to try and increase inflammation (jump-start it) to a point at which the inflammatory process will run its normal course and the injury can progress. |
| **FRICTION MASSAGE Keys to Applying** | Can be applied in a circular or transverse fashion. If circular, thumbs work in circular motion. If transverse, thumbs stroke tissue from opposite directions. Should be applied across fibers when treating a ligament or tendon. When treating scar tissue which has less regular organization, direction may be alternated or applied in a circular motion. ***47-49/73*** |
| **FRICTION MASSAGE Rules & Position** | Can use elbow on large muscles. Place muscle in relaxed position. Apply sufficient pressure to reach deep into tissue. Do not use on acute injury. Usually is painful. May be followed by stretching to increase ROM. |
| **TRANSVERSE FRICTION MASSAGE** | Technique for treating chronic tendon inflammation. **Purpose:** To increase inflammatory response to progress healing process. Use strong pressure in perpendicular direction to fibers for 7 to 10 minutes every other day. ***48-51/73*** |

## CONNECTIVE TISSUE MASSAGE (BINDEGE WEBS MASSAGE)

| OPERATIONAL FIELD | COMPLETE SCIENTIFIC & CLINICAL SPECIFICATIONS |
|---|---|
| **CONNECTIVE TISSUE MASSAGE Overview** | Stroking technique carried out in layers of connective tissue on body surface. Abnormal tension in one part of tissue is reflected in other parts. Stroking produces a relaxation of muscular tension and a prickling warmth in area. Used mostly in Europe. |

| OPERATIONAL FIELD | COMPLETE SCIENTIFIC & CLINICAL SPECIFICATIONS |
|---|---|
| **CONNECTIVE TISSUE MASSAGE Rules & Stroke** | Patient is usually in sitting position. Basic stroke of pulling performed with tips, or pads, of the middle and ring fingers of either hand. Stroking technique characterized by a tangential pull on skin and subcutaneous tissues away from fascia. Technique causes sharp pain in tissue. |
| **CONNECTIVE TISSUE MASSAGE Parameters** | No lubricant is used. Treatments last about 15 to 25 minutes. After 15 treatments 2–3 times per week, there should be a rest period of 4 weeks. Connective tissue massage must be learned and performed initially under direct supervision of someone who has been taught these highly specialized techniques. |
| **CONNECTIVE TISSUE MASSAGE Indications** | • Scars on the skin.<br>• Fractures and arthritis in bones and joints.<br>• Low back pain.<br>• Varicose symptoms, thrombophlebitis, hemorrhoids, edema in blood and lymph.<br>• Raynaud's disease, intermittent claudication, frostbite.<br>• Myocardial dysfunctions, respiratory disturbances.<br>• Intestinal disorders, ulcers, hepatitis, amenorrhea, dysmenorrhea, genital infantilism, Parkinson's disease, migraines. |

## TRIGGER POINT MASSAGE

| CLINICAL DIMENSION | UNABRIDGED ANATOMY, DIAGNOSIS & TREATMENT PROTOCOLS |
|---|---|
| **TRIGGER POINT MASSAGE Anatomical Sites** | Myofascial trigger points found in: Skeletal muscle and tendons, Myofascia, Ligaments surrounding joints, Periosteum. May be activated and become painful due to some trauma to muscle occurring either from direct trauma or from overuse. ***53-56/73*** |
| **TRIGGER POINT MASSAGE Neuro-Mechanism** | Pain results from inflammatory response. Pain usually referred to areas which follow a specific pattern. Stimulation of these points has been demonstrated to result in pain relief. Acupressure points and myofascial trigger points are similar. ***57-59/73*** |
| **TRIGGER POINT MASSAGE Classification** | **Latent trigger points:** Don't cause spontaneous pain. May restrict movement.<br>**Active trigger points:** Causes pain at rest. Tender to palpation with referred pain. |
| **TRIGGER POINT MASSAGE Identification** | Patient has persistent regional pain resulting in decreased ROM. Hypersensitive nodules palpation results in pain in the area and radiation of pain. Contracting involved muscle increases pain. Firm pressure usually elicits "jump sign". |
| **TRIGGER POINT MASSAGE Acupressure Overlap** | Acupressure and trigger point massage are very similar. Independently discovered but rely on similar underlying neural mechanisms. Treatment effectiveness may be result of intense, low-frequency stimulation of trigger points and release of β-endorphins. |
| **TRIGGER POINT MASSAGE TECHNIQUES** | Locate points from chart. Use fingers or elbow to do small friction-like circular motions. Amount of pressure applied should be intense and painful. Patient reports a dulling or numbing effect. Treatment times range from 1–5 min at several points. |

## STRAIN-COUNTERSTRAIN & ADVANCED RELEASE SYSTEMS

| ADVANCED SYSTEM | LITERAL TEXT PROTOCOLS, CLINICAL GUIDELINES & RATIONALES |
|---|---|
| **STRAIN-COUNTERSTRAIN Protocols** | Approach used to decrease muscle tension. Passive technique that places body in position of greatest comfort → pain relief. Athletic trainer locates a trigger point corresponding to point of dysfunction. Tend to be located deep in tendons, fascia & muscles. Tense, tender, edematous spots. Clinician monitors tension and pain of tender point while patient is moved into position of comfort. Often involves shortening of muscle/tissue. Tender spot will no longer be painful in this position. After 90 seconds pain and point should be cleared or reduced. Patient is then returned to resting position and should note change in symptoms associated with trigger point. |
| **STRAIN-COUNTERSTRAIN Rationale** | **Physiological rationale - Stretch reflex:** Muscle is placed on slack, reducing muscle spindle input. Facilitates relaxation and hence decrease tension and pain. |
| **POSITIONAL RELEASE THERAPY** | Based on strain-counterstrain. **Difference:** Use of a facilitating force. Follows the same steps as strain-counterstrain. Incorporates maintained compression on tender point. Suggested that maintaining contact exerts a therapeutic effect. |
| **ACTIVE RELEASE THERAPY** | Used to correct soft-tissue problems in: Muscle, Tendons, Fascia. Fibrotic adhesions due to: Acute injury, Pressure/tension injuries, Repetitive overuse injuries. Deep tissue technique. Clinician identifies the area and traps the affected muscle by applying pressure. Patient then actively elongates the muscle. Repeated 3–5 times. Patient must follow stretching, activity modification and exercise instructions. |
| **MYOFASCIAL RELEASE Overview & Treatment** | Has also been referred to as soft tissue mobilization. Group of stretching techniques used to relieve soft tissue from abnormal grip of tight fascia. Myofascial restrictions are unpredictable and may occur in many different planes and directions. Treatment is on localizing restriction and moving into the direction of the restriction. Soft tissue mobilization technique. Myofascial manipulation relies heavily on experience of clinician. Focuses on large treatment areas. Can have significant impact on joint mobility. Massage occurs through the restriction. With improvements in extensibility of tissue, stretching should be incorporated. Strengthening is also recommended to enhance neuromuscular re-education. Postural re-education may help ensure maintenance of less restricted movement patterns. Acute cases tend to resolve after a few treatments; while longer conditions require additional treatment. |
| **GRASTON TECHNIQUE** | Instrument-assisted soft tissue mobilization used to breakdown scar tissue and fascial restrictions. Uses handheld stainless steel instruments to: Scan, Locate, Treat injured tissues. Clinician will palpate painful area. Instruments help to magnify existing restrictions. Instruments allow for precise pressure application to break up scar tissue. Helps to relieve pain and restore function. Specially designed lubricant is designed to ensure that instruments glide over skin. Utilizes multi-directional cross-friction massage. Creates trauma and inflammatory response. Initiates and promotes healing process. Rehab activities and modalities should be used in conjunction. |